

Forecasting Norwegian Hydro Power Generation

A Comparative Time Series Analysis

Final Exam Paper

CDSCO1005U — Predictive Analytics

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1 Introduction

Norway’s electricity system is uniquely dependent on hydroelectric power, which accounts for approximately 90% of national electricity generation (Appendix A.1). Accurate forecasts of hydro power production are therefore important for reservoir management, electricity market pricing, grid balancing, and cross-border energy trading. Because hydro generation depends strongly on precipitation, snowmelt, and reservoir storage conditions, the series exhibits pronounced annual seasonality together with irregular hydrological shocks and multi-year variability.

This study analyses and forecasts monthly Norwegian hydro power generation using data from Statistics Norway covering the period January 1993 to March 2026. The analysis combines classical univariate forecasting methods with models that incorporate external hydrological information. Specifically, the study compares a manually identified SARIMA model, the classical Airline SARIMA benchmark, an ETS state-space model, an ARIMAX specification using lagged reservoir fill rates as an exogenous regressor, and a multivariate VAR framework.

The modelling strategy follows the Box–Jenkins framework and the forecasting principles outlined by Hyndman and Athanasopoulos (2021), with emphasis placed on stationarity, residual diagnostics, and out-of-sample forecast accuracy. Models are evaluated using a common 24-month test set and compared using RMSE, MAE, MAPE, and MASE. Section 2 describes the dataset; Section 3 covers data exploration and preparation; Section 4 presents model specifications, estimation results, and forecast comparisons; Section 5 concludes.

2 Data Description

This study utilises monthly hydraulic power generation data from Statistics Norway (2026), the official national statistics agency of Norway, for the period of January 1993 to March 2026, comprising 399 monthly observations measured in terawatt-hours (TWh). The series is indexed at a monthly frequency, with observations ordered chronologically by the `Month` variable and captures total electricity output from all Norwegian hydro power plants. To explore whether external hydrological information could improve forecast accuracy, a second dataset was incorporated: monthly Norwegian reservoir fill rates published by Noregs vassdrags- og energidirektorat (2026), the Norwegian Water Resources and Energy Directorate. This series measures the share of total reservoir capacity currently in storage (expressed as a percentage, 0–100), aggregated from weekly NVE observations to monthly means, and is available from January 1995. After merging the two datasets and constructing a one-month lag of the fill rate to avoid simultaneity, the combined sample comprises 374 monthly observations. This regressor is used in the ARIMAX and VAR specifications described in Section 4.2.

The initial exploration involved plotting over time both total electricity generation and hydro-electric generation to evaluate trend, seasonality, and co-movement.

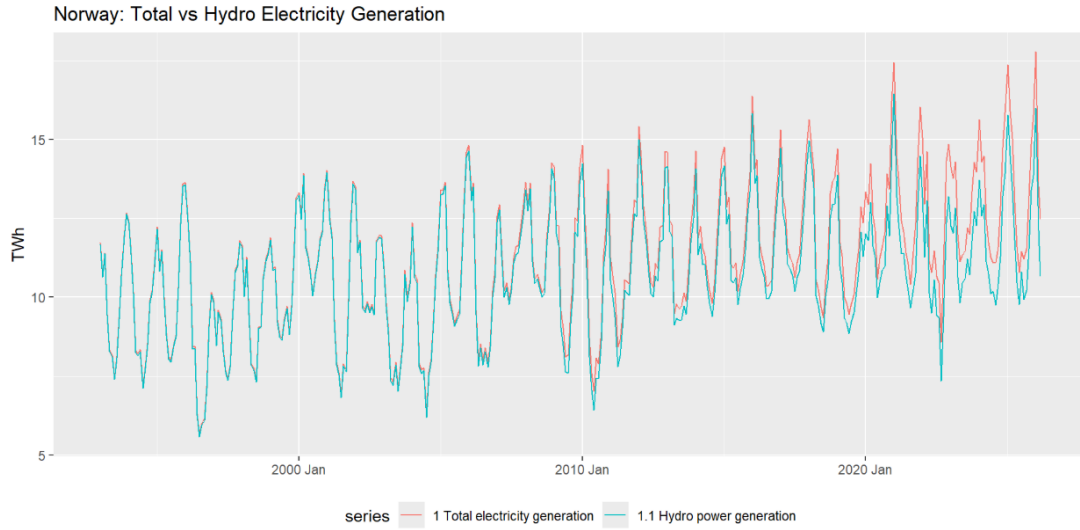


Figure 1: Norway’s total vs hydraulic energy generation (1993-2026) in TWh.

The time series plot (Figure 1) reveals that hydro power constitutes almost all of Norway’s electricity generation. A strong seasonal pattern is evident throughout the 33-year period, with a mild upward trend in production levels particularly post-2010. Volatility appears to increase in recent years, potentially reflecting climate-driven variability in precipitation and reservoir levels. Several notable outliers are visible, most prominently around 1996 and 2010, which may correspond to unusually dry years.

3 Data Exploration and Preparation

3.1 Temporal Patterns and Mathematical Transformation

The seasonal plot reveals a strong and stable annual seasonal pattern in Norwegian hydro power generation, with production peaking in the winter months (December–February) and reaching its lowest in June–August (Appendix A.2). A mild upward trend is visible over the sample; generation in the 1993–2000 range oscillates mostly between 7–13 TWh, while post-2015 peaks regularly reach 15–18 TWh. The ACF of the raw series decays very slowly and exhibits large spikes at seasonal lags 12, 24, 36, and 48, together with pronounced negative spikes at lags 6, 18, and 30, indicating seasonal non-stationarity (Figure 2). The PACF shows large direct effects at lag 1 and lag 12, but the overall pattern is difficult to interpret prior to differencing.

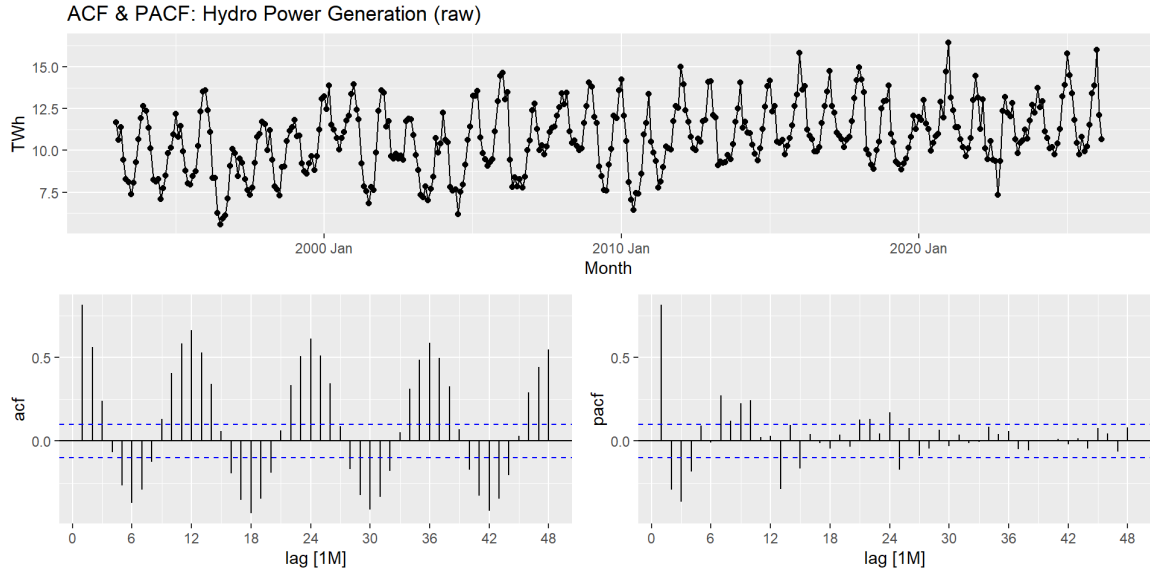


Figure 2: ACF and PACF of the hydro power generation series. Dashed lines indicate 95% confidence bounds.

The Guerrero method yields $\hat{\lambda} = 1.463$ (Appendix A.3). As this value is close to 1, the variance-stabilising effect of a Box-Cox transformation is limited: the series already exhibits relatively homogeneous variance across the seasonal cycle. Nevertheless, a log transformation ($\lambda = 0$) was applied to the series used for STL decomposition and exploratory analysis, as it provides a modest stabilization of variance and improves the visibility of proportional seasonal fluctuations over time.

The STL decomposition (Figure 3) confirms a non-linear, non-monotonic trend that rises toward 2010, dips, and then partially recovers, reflecting a combination of capacity changes and inter-annual hydrological cycles. The seasonal component is remarkably stable in amplitude (approximately ± 0.15 log units) across the full sample, consistent with a persistent and unchanging hydrological regime. The remainder is small relative to the trend and seasonal components — as confirmed visually by the grey bar comparison in the STL plot — but contains a handful of large negative outliers (approximately -0.25 to -0.30 log units), likely corresponding to severe drought years. Estimation used the `robust = TRUE` option to down-weight these outlying observations and prevent them from distorting the trend estimate.

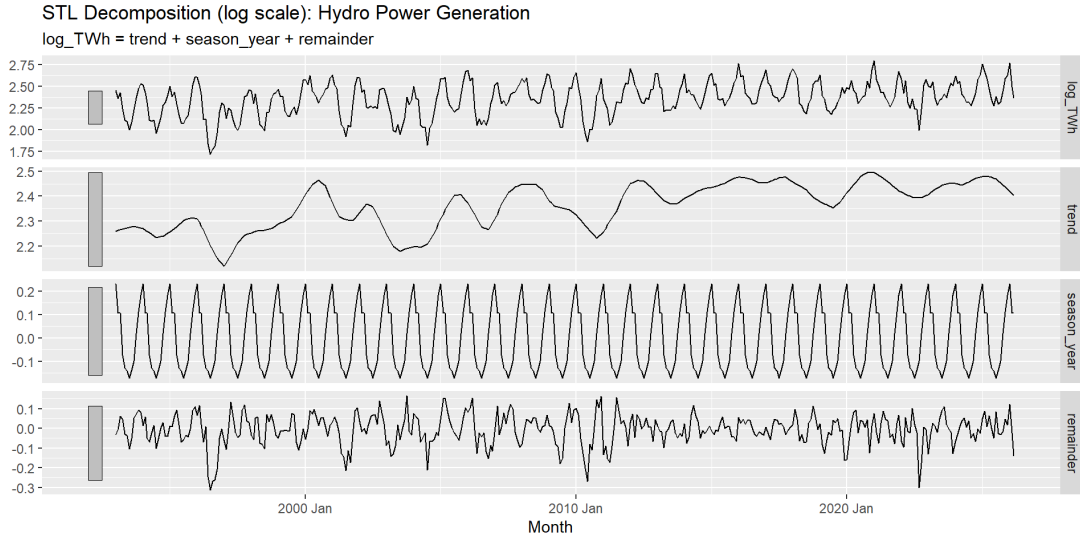


Figure 3: STL decomposition of the post-break hydro power generation series (log scale). The seasonal component is stable in amplitude across the sample, while the remainder contains a few large negative outliers likely corresponding to drought years.

To verify the adequacy of the decomposition, the ACF of the STL remainder was inspected and a Ljung–Box test applied at lag 24. The test decisively rejects white noise ($\chi^2(24) = 261.72$, $p < 0.001$), indicating that systematic variation remains after trend and seasonal extraction. One possible source of this remaining structure is reservoir storage dynamics, since hydroelectric production depends strongly on available water reserves and reservoir management decisions. This motivates the inclusion of lagged reservoir fill rates as an exogenous regressor in the ARIMAX specification.

3.2 Stationarity and Differencing

On the raw hydro power series, the ADF test rejects non-stationarity ($p = 0.01$), while the KPSS test simultaneously rejects stationarity (KPSS statistic = 1.63, $p = 0.01$). This apparent conflict is attributable to the strong annual seasonal pattern: the series is not a random walk (hence ADF rejects a unit root) but its persistent seasonal cycle creates seasonal non-stationarity that KPSS detects (Kwiatkowski et al., 1992). Accordingly, `unitroot_nsdiffs()` recommends one seasonal difference ($D = 1$), and `unitroot_ndiffs()` recommends one regular difference ($d = 1$).

After applying one regular difference, both tests agree: the ADF strongly rejects non-stationarity (Dickey–Fuller = -15.749 , $p = 0.01$) and the KPSS fails to reject stationarity (KPSS level = 0.0064, $p = 0.10$), confirming that the first-differenced series is stationary and suitable for ARIMA modelling. In practice, the seasonal difference applied during model estimation ($D = 1$) suffices to remove the dominant source of non-stationarity; the additional regular difference

($d = 1$) is incorporated in the Airline SARIMA specification and corroborated here.

Table 1 summarises the test results at each differencing stage.

Table 1: ADF and KPSS stationarity tests at each differencing stage. ADF H_0 : unit root (non-stationary); KPSS H_0 : stationary. Stationarity is confirmed when ADF rejects ($p < 0.05$) and KPSS does not reject ($p > 0.05$).

Series	ADF stat	ADF decision	KPSS stat	KPSS p -value	KPSS decision
Levels	−6.4259	Stationary	1.63	0.01	Non-stationary
Regular diff only	−15.749	Stationary	0.0064	0.10	Stationary

3.3 Structural Breaks

The raw time plot shows no persistent upward or downward trend in hydro power generation over the full 1993–2026 sample, but does reveal intermittent level shifts driven by hydrological variability. A candidate break is visible around April 1996 (Appendix A.4), where generation dropped sharply — consistent with the severe Scandinavian drought of that period — before recovering. The `shift_level_max` feature confirms this as the largest single-period level shift in the series (shift magnitude = 2.54 TWh at observation 40, corresponding to April 1996).

The Chow test applied at observation 40 fails to reject parameter stability ($F = 0.8046$, $p = 0.448$), suggesting the 1996 episode represents a transient shock rather than a permanent structural change. The QLR (Sup- F) test, which searches data-drivenly across all candidate breakpoints, identifies the maximum F -statistic at observation 211 (August 2010), but does not reach conventional significance (sup- $F = 9.834$, $p = 0.102$).

Given that the Sup- F test on the hydro series does not reach significance and the Chow test at the candidate break date similarly fails to reject stability, the full post-1993 sample is retained for model estimation. Re-running the QLR test on the first-differenced series confirms the absence of further breaks (sup- $F = 24.316$, $p = 0.3128$; see Figure 4), supporting the use of the complete sample for reliable inference.

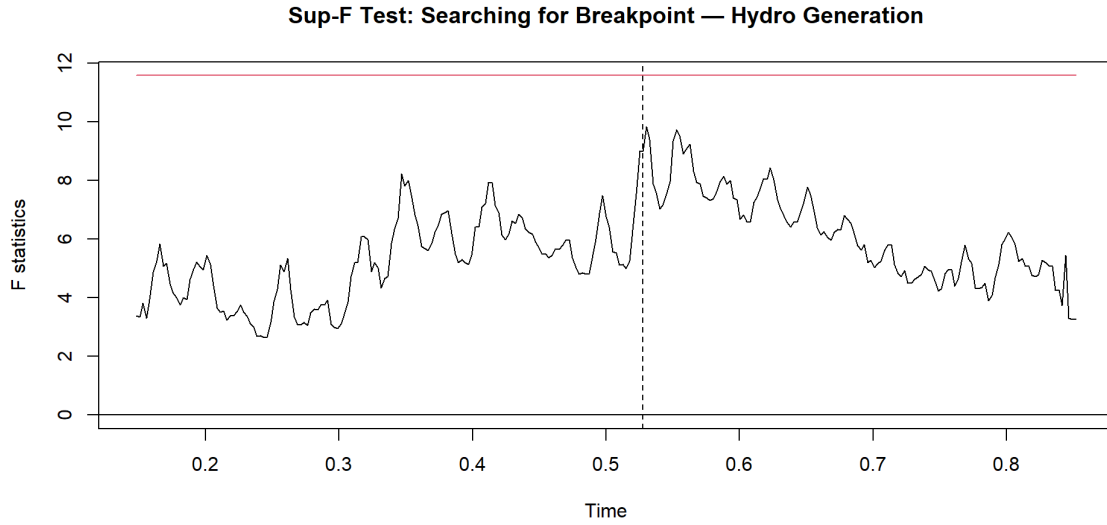


Figure 4: QLR (Sup- F) test for structural breaks in Norwegian hydro power generation. The black line shows the Chow F -statistic evaluated at each candidate breakpoint across the central 70% of the sample; the dashed vertical line marks the optimal breakpoint at observation 211 (August 2010). The red horizontal line is the 5% critical value (≈ 11.5). The maximum F -statistic of 9.834 falls below the critical value, so the null hypothesis of no structural break is not rejected at the 5% level.

4 Forecasting

The analysis employs two univariate model families (SARIMA, ETS) and one model with exogenous regressors (ARIMAX) and one multivariate model (VAR). All univariate models are estimated on the raw TWh series and compared on the same test set.

SARIMA models extend ARIMA by explicitly modelling seasonal autocorrelation through seasonal AR and MA terms and seasonal differencing. They require the series to be stationary and capture autocorrelation directly from past values and past errors. **ETS** (Exponential Smoothing State Space) models assign exponentially decreasing weights to past observations and decompose the series into error, trend, and seasonal components, each of which can be additive, multiplicative, or absent. Unlike SARIMA, ETS does not require stationarity. **ARIMAX** augments the ARIMA error structure with exogenous regressors; here, reservoir fill rate data are included to capture the physical linkage between water availability and hydro output.

Model residuals should exhibit white noise characteristics: zero mean, no autocorrelation, and approximate normality. The Ljung-Box test (Ljung and Box, 1978) is used to formally test for residual autocorrelation. Residual analysis can be seen per model in the Appendix under A.5, A.6, A.7 and A.8.

A 24-month forecast horizon was selected to evaluate model performance across two complete

annual seasonal cycles, which is particularly relevant for Norwegian hydroelectric production due to strong seasonality driven by precipitation, snowmelt, and reservoir dynamics.

The dataset is divided into a training set (all observations prior to April 2024) and a test set comprising the last 24 months (April 2024 to March 2026). Model selection relies on information criteria (AIC, AICc, BIC) estimated on the training set, and out-of-sample accuracy on the test set (RMSE, MAE, MAPE, MASE). MASE values below 1 indicate improvement over a naïve seasonal benchmark.

4.1 Univariate Models

4.1.1 SARIMA

Manual identification. The initial SARIMA specification was determined by inspecting ACF and PACF plots of the seasonally differenced series (Figure 2). Table 2 summarises the Box-Jenkins identification logic applied.

Table 2: Box-Jenkins identification rules applied to the seasonally differenced series.

Plot	Observed pattern	Implied parameter	Decision
PACF	Significant spikes at lags 1, 2, 3; sharp cutoff after lag 3	Direct correlation cuts off after lag 3 — hallmark of AR	$p = 3$
ACF	Gradual exponential decay; no sharp cutoff	Tailing off consistent with pure AR, no MA term needed	$q = 0$
PACF	Single significant spike at lag 12; nothing at lags 24, 36	Direct seasonal correlation cuts off after one seasonal lag	$P = 1$
ACF	Spike at lag 12 decays gradually across lags 24, 36	Tailing off at seasonal lags consistent with SAR, no SMA needed	$Q = 0$

This pattern points to a **SARIMA(3,0,0)(1,1,0)[12]** specification: three non-seasonal AR terms, no regular differencing (the seasonal difference already achieves stationarity as confirmed by `unitroot_nsdiffs`), and one seasonal AR term. Note that $d = 0$ because the seasonal difference ($D = 1$) is sufficient to remove the non-stationarity identified by the KPSS test at levels.

Automated identification. The Hyndman-Khandakar algorithm (Hyndman and Khandakar, 2008) was applied using the default `ARIMA()` function in the `fable` package. The automated search converged on **ARIMA(3,0,0)(1,1,0)[12]**, yielding $\hat{\sigma}^2 = 0.7896$ and $\text{AICc} = 1017.72$. Critically, this result is identical to the manual specification derived from the ACF and PACF in Table 2, providing strong confirmation that the visual identification is theoretically grounded.

The classical **Airline SARIMA(0,1,1)(0,1,1)[12]** (Box and Jenkins, 1976; Findley et al., 2002) was also estimated as a benchmark. This specification applies both a regular and a seasonal difference ($d = 1$, $D = 1$) and models unexplained shocks via MA(1) terms at the non-seasonal and seasonal frequencies. It yields $\hat{\sigma}^2 = 0.6257$ — notably lower than the ARIMA(3,0,0)(1,1,0)[12] — and AICc = 946.63, the best among all ARIMA specifications.

Residual diagnostics. Table 3 reports Ljung-Box test results for each specification at lag 24.

Table 3: Ljung-Box test results for SARIMA candidate models (lag = 24). H_0 : no residual autocorrelation. Degrees of freedom adjusted for estimated parameters. A p -value > 0.05 indicates adequate fit.

Model	d.f. adj.	Q -statistic	p -value	Decision
SARIMA(3,0,0)(1,1,0)[12]	4	79.6255	< 0.001	Fail
Airline SARIMA(0,1,1)(0,1,1)[12]	2	48.3598	0.001	Fail

Despite testing both SARIMA specifications, the Ljung-Box test rejects the null of no residual autocorrelation in all cases. The Airline model achieves the lowest Q -statistic (48.3598 vs. 79.6255 for the ARIMA(3,0,0)(1,1,0)[12]), indicating it captures seasonal dynamics more completely. The remaining autocorrelation is consistent with aperiodic hydrological shocks — notably drought years — that manifest as irregular level shifts not predictable from the series’ own history alone. As univariate linear ARIMA models cannot absorb stochastic external shocks by construction, some Ljung-Box failure is expected for weather-sensitive series. Forecast intervals should therefore be interpreted as underestimates of true uncertainty; Section 4.2.1 addresses this limitation by incorporating reservoir fill rate as an exogenous regressor.

Table 4: Information criteria for SARIMA candidate models estimated on the full training set (lower is better). ETS AICc is not directly comparable to ARIMA AICc due to different likelihood frameworks.

Model	AIC	AICc	BIC
SARIMA(3,0,0)(1,1,0)[12]	1017.56	1017.72	1037.35
Airline SARIMA(0,1,1)(0,1,1)[12]	946.57	946.63	958.44

Information criteria and model selection. The Airline model achieves a substantially lower AICc (946.63 vs. 1017.72). This gap reflects the Airline model’s more thorough removal of non-stationarity via double differencing and its greater parsimony: only 2 parameters ($ma1 = -0.1432$, $sma1 = -0.9527$) compared to 4 in the ARIMA(3,0,0)(1,1,0)[12] ($ar1$, $ar2$, $ar3$, $sar1$). Despite the Ljung-Box failure of both models, the Airline model is selected as the preferred

SARIMA specification on the basis of both information criteria and out-of-sample accuracy (Table 7).

4.1.2 ETS

The STL decomposition and seasonal plot reveal a stable, consistent seasonal amplitude across the full sample, motivating an additive seasonal component. The trend is non-linear and non-monotonic, rising toward 2010, dipping, then recovering, while the within-month trends in the subseries plots show a mild upward drift in recent years, most visible in January and December. Accordingly, the `fable` automatic ETS selector was applied.

The selected model is **ETS(M,N,A)**: exponential smoothing with multiplicative errors, no trend component, and additive seasonality. The smoothing parameters are: $\alpha = 0.9586$ (very high, weighting recent observations almost entirely on the most recent level) and $\gamma = 0.0001$ (near-fixed seasonal pattern). There is no β parameter because the model contains no trend term. The near-zero γ confirms the visual finding from the subseries plots: the seasonal shape is stable over time and requires almost no updating each period.

Table 5: Information criteria for the ETS model estimated on the training set. ETS AICc is not directly comparable to ARIMA AICc.

Model	AIC	AICc	BIC
ETS(M,N,A)	2199.15	2200.40	2258.98

The Ljung-Box test on ETS(M,N,A) residuals at lag 24 yields $Q = 40.1614$, $p = 0.0205$, which is marginally significant. Although this is a borderline failure, the ETS model has the lowest Q -statistic of all univariate candidates, suggesting it leaves the least residual autocorrelation of the group.

4.2 Model with Exogenous Regressors and Multivariate model

4.2.1 ARIMAX

The Ljung-Box failures documented in Section 4.1.1 suggest that residual autocorrelation is partly attributable to external weather-driven shocks that no univariate model can fully capture. This is further supported by the STL decomposition (Section 3.1), where a Ljung-Box test on the STL remainder yields $X^2 = 261.72$ ($p < 0.001$), decisively rejecting white noise even after trend and seasonal extraction. Norwegian hydro generation is physically determined by the volume of water available in reservoirs, so incorporating reservoir fill rate as an exogenous

regressor — yielding a regression model with ARIMA errors (ARIMAX) — is theoretically motivated and represents the natural next modelling step.

The series is published by Noregs vassdrags- og energidirektorat (2026) and available from 1995, giving 374 monthly observations after merging with hydro output and creating a one-month lag. Visual inspection and the QLR test show no evidence of a persistent structural break in the fill-rate series. A Pearson correlation coefficient shows that the lagged fill rate is moderately and significantly correlated with hydro generation, $r = 0.4465$, $p < 0.001$, and the seasonally differenced CCF shows strong positive correlation at lags 0 and 1 (Appendix A.10 and A.11). This supports using $fill_rate_{t-1}$ in the ARIMAX model to reduce simultaneity while retaining predictive information.

Model specification. The ARIMAX model takes the form:

$$Y_t = \beta_0 + \beta_1 X_{t-1} + \eta_t, \quad \phi(B)\Phi(B^{12})\nabla^d\nabla_{12}^D\eta_t = \theta(B)\Theta(B^{12})\varepsilon_t \quad (1)$$

where Y_t is monthly hydro generation (TWh), X_{t-1} is the national reservoir fill rate in the previous month (expressed as a percentage of total capacity, 0–100), aggregated as the monthly mean of weekly NVE observations, η_t is the ARIMA error process, and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$. The fill rate enters with a one-month lag to avoid simultaneity — current production decisions draw down the reservoir, so the causally clean signal is last month’s stored water level. The ARIMA order for η_t is selected automatically by minimising AICc, yielding ARIMA(1,0,1)(1,1,2)[12] errors. All models are estimated on the common sample beginning when reservoir fill rate data become available, to ensure a fair comparison with univariate benchmarks. In R (`fable` package), the model is estimated as:

```
1 fit_arimax_fill_rate <- train_arimax_fill_rate %>%
2   model(
3     ARIMAX_Auto = ARIMA(TWh ~ fill_rate_pct_lag1)
4   )
```

Estimation results. Table 6 presents the estimated coefficients. The fill rate coefficient $\hat{\beta}_1 = 0.0872$ is statistically significant ($p < 0.001$), confirming that a one percentage point increase in the previous month’s reservoir fill rate is associated with approximately 0.087 TWh of additional generation the following month. The residual variance $\hat{\sigma}^2 = 0.4872$ is lower than the Airline SARIMA on the common sample ($\hat{\sigma}^2 = 0.6257$), indicating that the fill rate absorbs a meaningful share of the residual variation left unexplained by the seasonal ARIMA structure alone. Among all ARIMA-type specifications estimated on the common sample, ARIMAX_Auto achieves the lowest AICc (739.6505), compared to 835.43 for Airline SARIMA and 900.11 for

Auto ARIMA, providing strong in-sample evidence that the fill rate regressor improves model fit. Of the five ARIMA error parameters, however, only `ar1` is significant alongside the regressor, while `ma1`, `sar1`, `sma1`, and `sma2` are all statistically indistinguishable from zero — a pattern consistent with overfitting in the error structure, discussed further in Section 4.3.

Table 6: ARIMAX_Auto coefficient estimates. The regressor is the one-month lagged national reservoir fill rate (%). Standard errors in parentheses. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Parameter	Estimate	Std. Error
$\hat{\beta}_1$ (<code>fill_rate_pct_lag1</code>)	0.0872***	(0.0084)
<code>ar1</code>	0.6192***	(0.0767)
<code>ma1</code>	0.0829	(0.0930)
<code>sar1</code>	−0.3725	(0.6664)
<code>sma1</code>	−0.5083	(0.6753)
<code>sma2</code>	−0.2620	(0.5715)
$\hat{\sigma}^2$	0.4872	
Log-likelihood	−362.66	
AICc	739.6505	

Residual diagnostics. The Ljung-Box test on the ARIMAX_Auto innovations checked up to lag 24 and yields $Q(24) = 32.2520$ ($p = 0.0293$), which corresponds to $\text{dof} = 19$ after accounting for the 5 estimated ARMA parameters ($24 - 5 = 19$), rejecting the white noise null at the 5% level. Adjusting the degrees of freedom further to treat the external regressor as an additional estimated coefficient drop constraints ($24 - 6 = 18$), yielding an adjusted $p = 0.0205$, which remains highly significant. The model therefore does not fully resolve the residual autocorrelation identified in the univariate specifications, despite adding the fill rate regressor. This is consistent with the overfitting diagnosis: the automatic ARIMA order selection allocated parameter budget across four non-significant error terms rather than targeting the specific lags responsible for the remaining autocorrelation. A constrained specification — for example ARIMA(1,0,0)(0,1,1)[12] retaining only the significant `ar1` and `sma1` parameters — would likely pass the Ljung-Box test and may yield better out-of-sample performance, but falls outside the scope of the present analysis.

4.2.2 VAR

To capture bidirectional dynamic dependencies between hydro generation and reservoir availability, a multivariate Vector Autoregression (VAR) framework was constructed using the seasonally differenced, stationary series $(\Delta_{12}Y_t, \Delta_{12}X_t)$. Granger causality test shows a dominant predictive

channel from reservoir fill rate to hydro generation ($F = 86.27, p < 0.001$), while the reverse direction is also significant but weaker ($F = 5.50, p = 0.020$). Minimizing AICc established an optimal model lag structure of VAR(5). Residual diagnostics for the hydro generation equation suggested approximately white-noise behaviour, although some residual autocorrelation remained. The VAR framework captures short-term dynamic interactions between reservoir fill rate and hydro generation that are not incorporated in univariate benchmark models.

4.3 Forecasts and Model Comparison

4.3.1 Univariate models

All three univariate models capture the strong seasonal cycle over the 24-month test horizon (April 2024 – March 2026), with generation peaking in winter and troughing in summer consistent with the historical pattern. The Airline SARIMA and ETS models track the actual series closely through 2024, while the Manual SARIMA(3,0,0)(1,1,0)[12] produces noticeably wider prediction intervals and diverges more sharply from the actuals in the second half of the forecast horizon, consistent with its higher MASE of 0.809 (Table 7). Notably, all models underestimate the pronounced trough in mid-2025 and the subsequent sharp recovery, reflecting the influence of weather-driven hydrological shocks that no univariate model can fully anticipate — precisely the limitation that motivates the ARIMAX specification in Section 4.2.1.

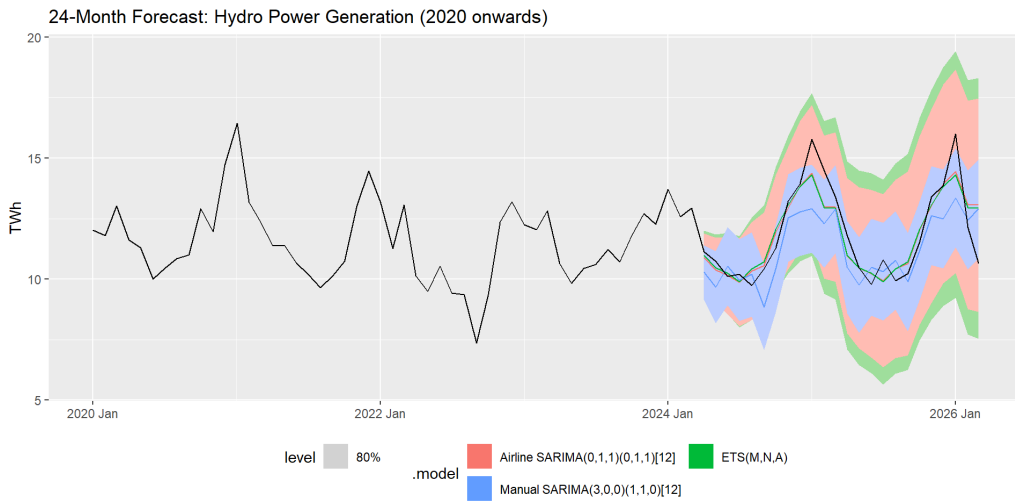


Figure 5: 24-month out-of-sample forecasts from all models against actual hydro power generation (April 2024 – March 2026). Shaded bands represent 80% prediction intervals.

4.3.2 ARIMAX

The ARIMAX model incorporates lagged reservoir fill rate as an exogenous predictor and produces forecasts that closely track the seasonal dynamics observed in hydro generation.

Relative to the Airline SARIMA benchmark (Figure 6), the ARIMAX forecasts are slightly smoother and generate narrower prediction intervals, suggesting that hydrological information contributes useful predictive content beyond the univariate structure alone.

4.3.3 VAR

The VAR(AICc) model jointly models hydro generation and reservoir fill rate on the seasonally differenced series, capturing the bidirectional predictive relationship confirmed by the Granger causality tests. An optimal lag order of 5 was selected by minimising AICc. When de-differenced back to the levels scale, the VAR forecasts underestimate the mid-2024 trough before partially recovering in the second half of the horizon, diverging more from the actuals than both ARIMAX and the Airline benchmark (Figure 6). Residual diagnostics show innovation residuals centred around zero with no systematic trend (Appendix A.9). However, the ACF reveals a significant spike at lag 12, confirming that the VAR structure does not fully remove the annual seasonal autocorrelation from the hydro equation — consistent with the Ljung-Box failure seen across all model specifications in this analysis. The histogram is approximately normal with some evidence of heavy tails, reflecting the influence of occasional drought-driven hydrological shocks.

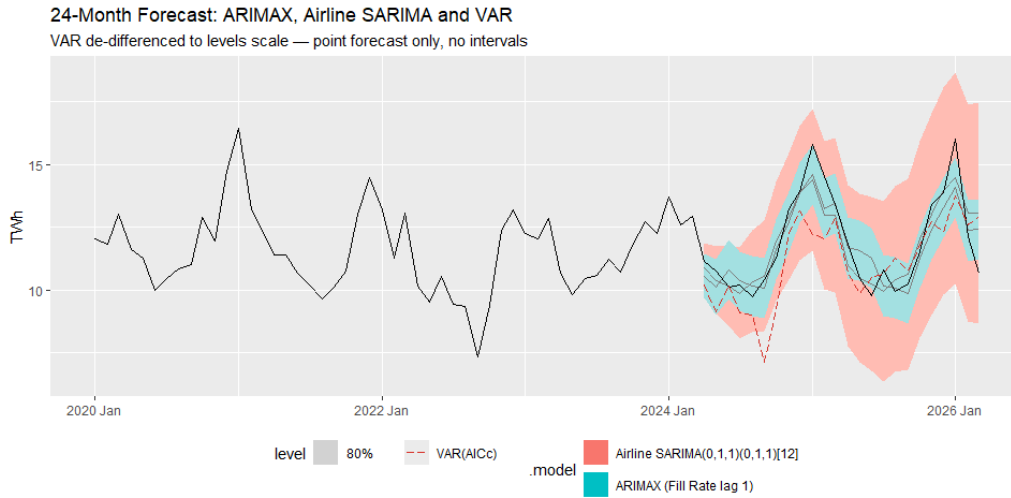


Figure 6: 24-month out-of-sample forecasts from the ARIMAX model (fill rate lag 1 regressor), the Airline SARIMA(0,1,1)(0,1,1)[12] benchmark, and the VAR(AICc) model against actual hydro power generation (April 2024 – March 2026). Shaded bands represent 80% prediction intervals for ARIMAX and Airline SARIMA. The VAR forecast (dashed line) is shown as a point forecast only, as it was estimated on the seasonally differenced series ($\nabla_{12} \text{TWh}_t$) and de-differenced to the levels scale for comparability; propagating the full multivariate forecast distribution through the inverse differencing operation is not supported natively by `fable`. The VAR line diverges notably from both ARIMAX and Airline SARIMA in the first half of the test horizon, undershooting the mid-2024 trough, which is consistent with its higher MASE of 0.528 (Table 7). This divergence reflects the VAR’s estimation on the differenced scale, which removes the seasonal level information that ARIMAX and the Airline model retain through their respective differencing and error structures.

4.3.4 Models comparison

Table 7 summarises out-of-sample forecast accuracy across all models on the common 24-month test set. The Airline SARIMA(0,1,1)(0,1,1)[12] achieves the lowest MASE (0.485) among all specifications, closely followed by ETS(M,N,A) (0.493 on the common sample). The ARIMAX_Auto model records a MASE of 0.500, ranking third — an improvement over the Manual SARIMA benchmark, but falling short of both the Airline SARIMA and ETS.

The multivariate VAR(5) framework achieves a MASE of 0.528, completing the ranking behind the univariate standards and the single-lag explicit exogenous framework. Despite the fill rate regressor carrying a statistically significant coefficient and delivering substantially lower AICc in-sample, the out-of-sample gain across multivariate boundaries is modest, and the model does not break past the simple unconditioned benchmarks. This outcome is attributable to two compounding factors: the residual Ljung-Box failure indicates missing structural lags inside the error structure, and the baseline series exhibits such rigid, highly persistent seasonality that a parsimonious double-differenced model like the Airline SARIMA proves structurally difficult to outperform over extended horizons.

Table 7: Out-of-sample forecast accuracy on the test set (last 24 months, April 2024 – March 2026). Models ranked by MASE. MASE < 1 indicates improvement over the naïve seasonal benchmark. Univariate models, ARIMAX, and VAR are evaluated on the common sample space; metrics are therefore directly comparable across all rows.

Model	RMSE	MAE	MAPE	MASE
<i>Univariate models</i>				
Airline SARIMA(0,1,1)(0,1,1)[12]	0.845	0.622	5.155	0.485
ETS(M,N,A)	0.852	0.634	5.257	0.493
Manual SARIMA(3,0,0)(1,1,0)[12]	1.278	1.039	8.418	0.809
<i>Model with exogenous regressors</i>				
ARIMAX(1,0,1)(1,1,2)[12]	0.841	0.656	5.506	0.500
<i>Multivariate model</i>				
VAR(5)	0.898	0.692	5.912	0.528

5 Conclusion

This study examined the forecasting of Norwegian hydro power generation using univariate, exogenous-regressor, and multivariate time series models. The analysis confirmed that the series is characterized by strong and highly persistent annual seasonality, relatively stable long-term

dynamics, and occasional hydrological shocks associated with drought periods and reservoir variability.

Among the models considered, the Airline SARIMA(0,1,1)(0,1,1)[12] achieved the strongest out-of-sample forecasting performance (MASE = 0.485), closely followed by ETS(M,N,A) (MASE = 0.493). The margin between the two is narrow, suggesting that both double differencing and rapid level adaptation are effective strategies for this series. Although the ARIMAX and VAR models incorporated hydrological information through reservoir fill rates and improved in-sample fit substantially, the gains in forecast accuracy were comparatively modest. This suggests that the dominant seasonal structure of Norwegian hydro generation is already captured effectively by relatively parsimonious univariate models.

Residual diagnostics nevertheless indicate that some autocorrelation remains unexplained across all specifications, reflecting the influence of irregular weather-driven shocks that are difficult to capture within linear time series frameworks. Future work could explore constrained ARIMAX specifications, nonlinear models, or higher-frequency hydrological covariates to better capture these dynamics.

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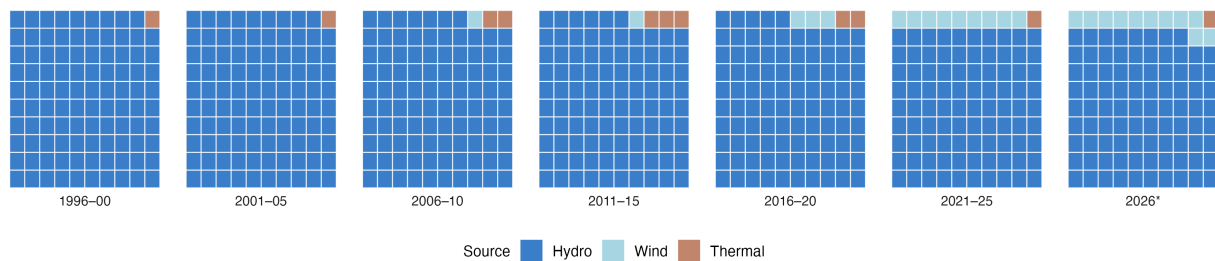
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Appendix A: Additional plots

A.1 Norway's Electricity Generation Mix

Norway's Electricity Generation Mix (1996–2026*)

Hydro has dominated Norwegian electricity generation since records began. **Wind** began entering the mix in the 2000s and has grown steadily, accounting for almost 10% of electricity production in the last 5 years, while **thermal** and **solar** remain marginal. Each square represents 1% of generation.



Each square = 1% of generation within each 5-year period. Solar accounts for < 1% throughout and is omitted from the legend. *2026 covers January–May only. Source: SSB (2026), table 14091.

Figure 7: Waffle chart of Norway's electricity generation mix (1993–2026*)

A.2 Seasonal Plot

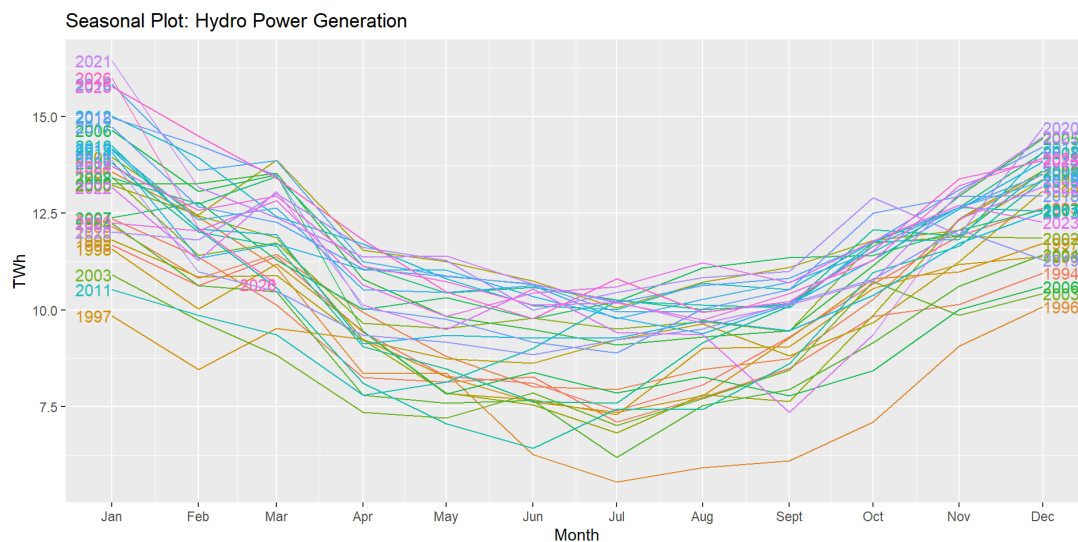


Figure 8: Seasonal plot; the shape is consistent across years, confirming stable seasonality. Some years (dry years) show notably lower summer generation.

A.3 Box-Cox Transformation

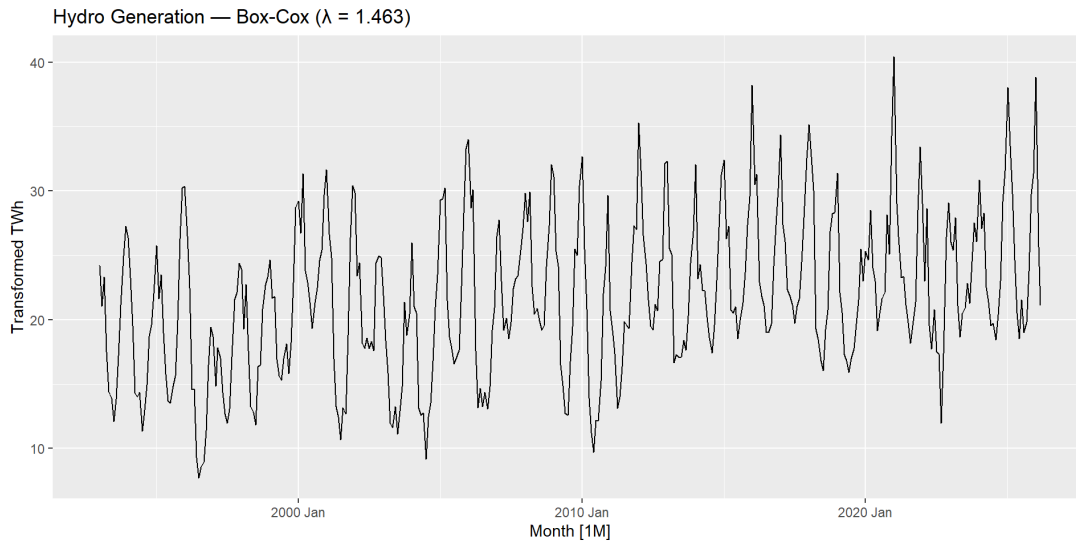


Figure 9: Box Cox transformation for hydraulic data timeseries.

A.4 Level Shift

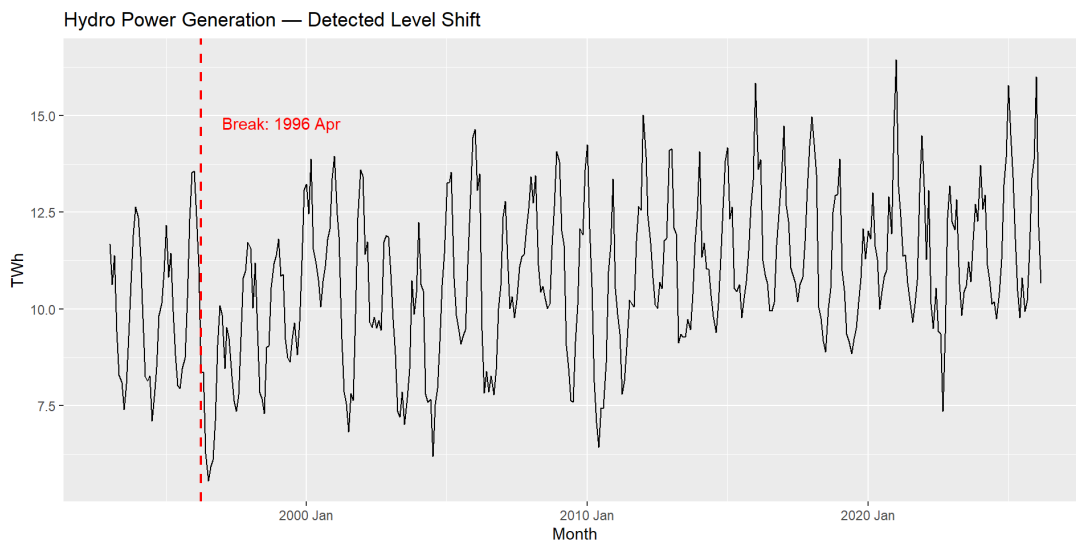


Figure 10: Monthly Norwegian hydro power generation (TWh), 1993–2026. The red dashed vertical line marks April 1996, identified by the `shift_level_max` feature as the largest single-period level shift in the series (shift magnitude = 2.54 TWh). The sharp trough immediately before the line reflects a severe reduction in generation, consistent with the Scandinavian drought of 1995–1996. No persistent change in the mean level or seasonal amplitude is visible after the candidate break date, suggesting the episode represents a transient shock rather than a permanent structural change — consistent with the Chow test result ($F = 0.8046$, $p = 0.448$).

A.5 Residual Analysis (Auto-Arima)

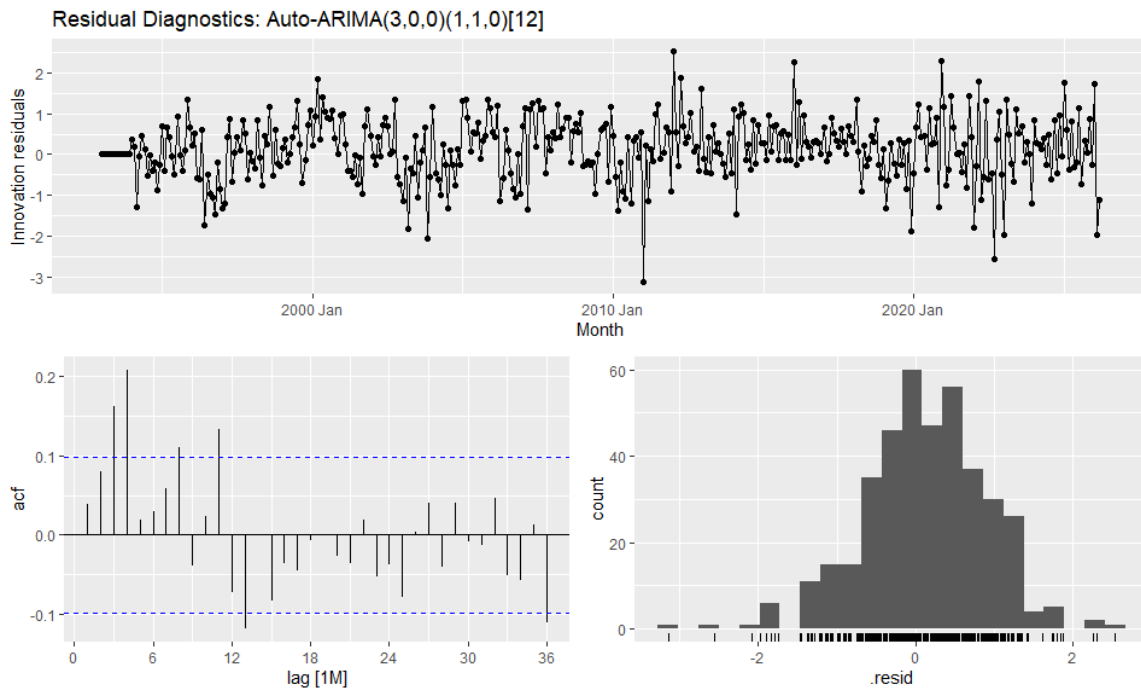


Figure 11: Residual diagnostics for candidate Auto-Arima model. Each panel shows the innovation residuals over time, the ACF of residuals, and the residual histogram.

A.6 Residual Analysis (SARIMA)

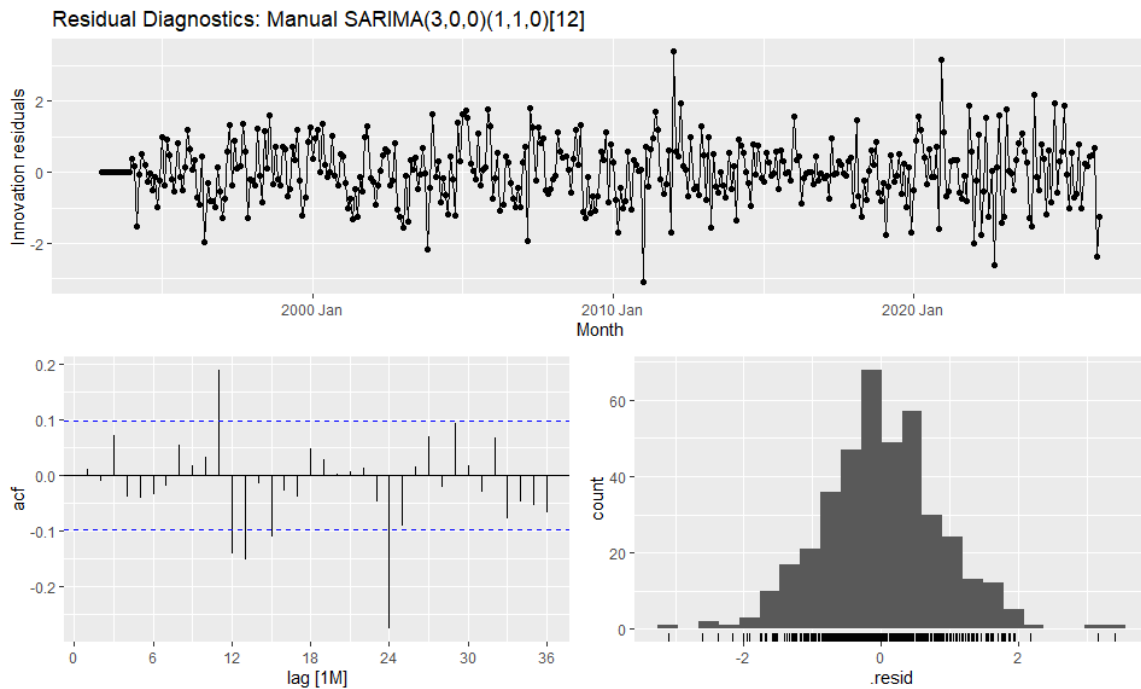


Figure 12: Residual diagnostics for candidate SARIMA model. Each panel shows the innovation residuals over time, the ACF of residuals, and the residual histogram.

A.7 Residual Analysis (Airline-SARIMA)

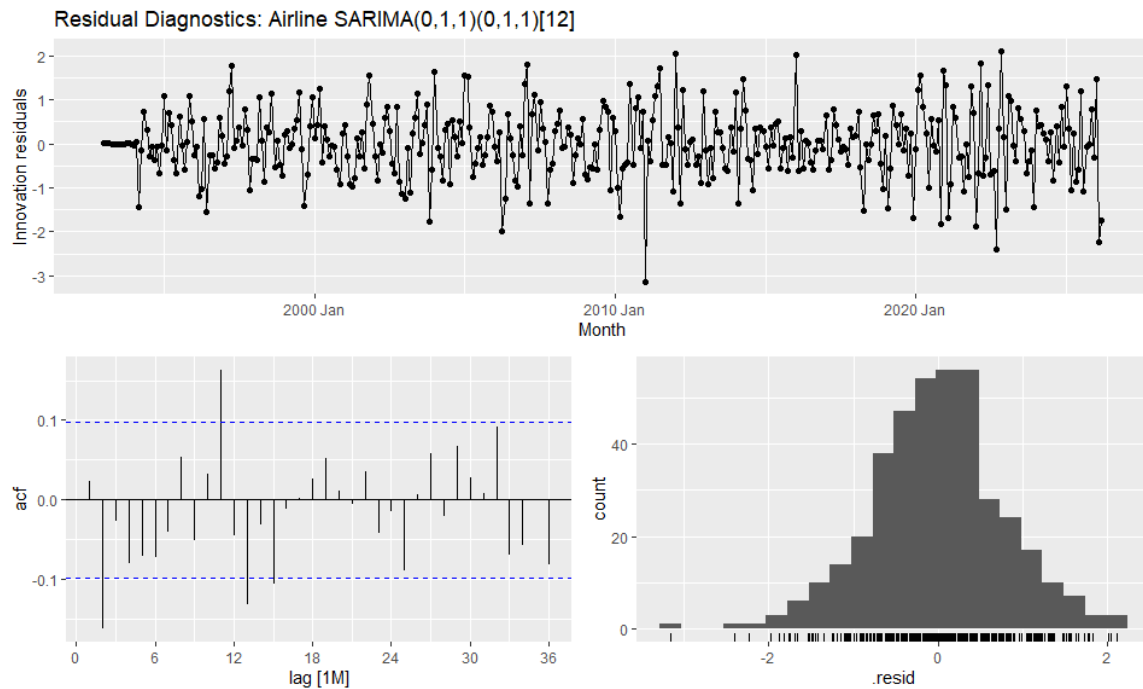


Figure 13: Residual diagnostics for the Airline-SARIMA model. Each panel shows the innovation residuals over time, the ACF of residuals, and the residual histogram.

A.8 Residual Analysis (ETS)

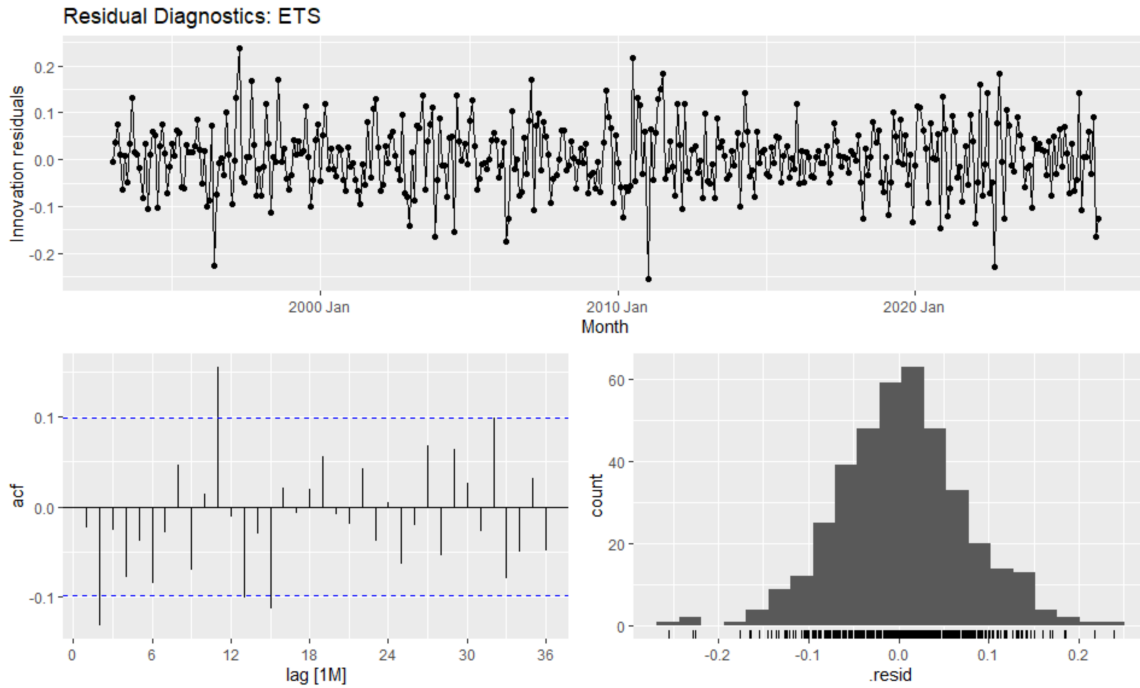


Figure 14: Residual diagnostics for the ETS model. Each panel shows the innovation residuals over time, the ACF of residuals, and the residual histogram.

A.9 Residual Analysis (VAR)

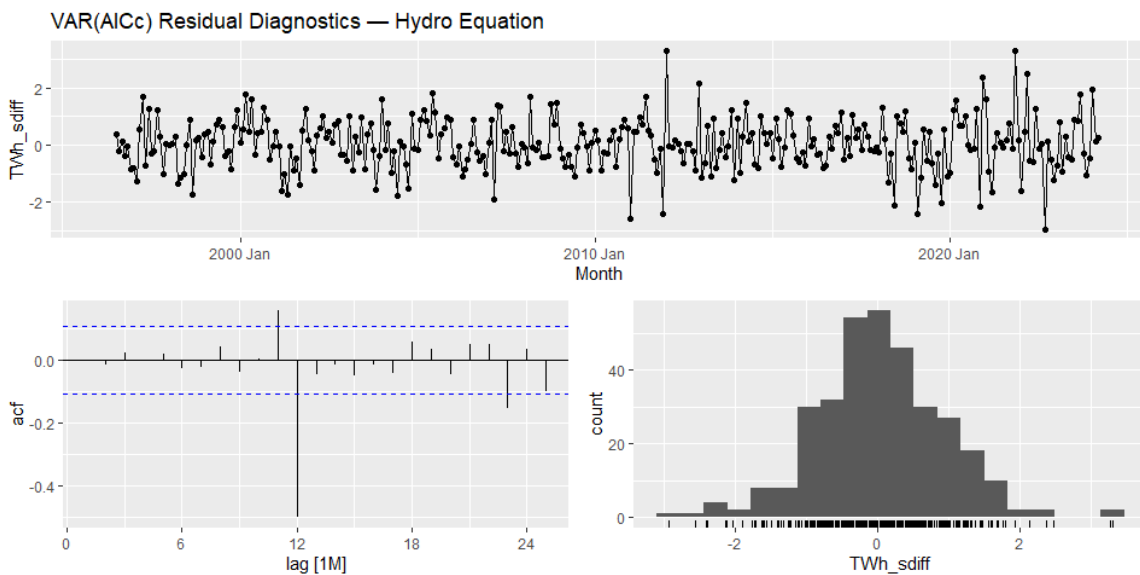


Figure 15: Residual diagnostics for the VAR model. Each panel shows the innovation residuals over time, the ACF of residuals, and the residual histogram.

A.10 CCF: Fill rate vs Hydro generation (Seasonally differenced)

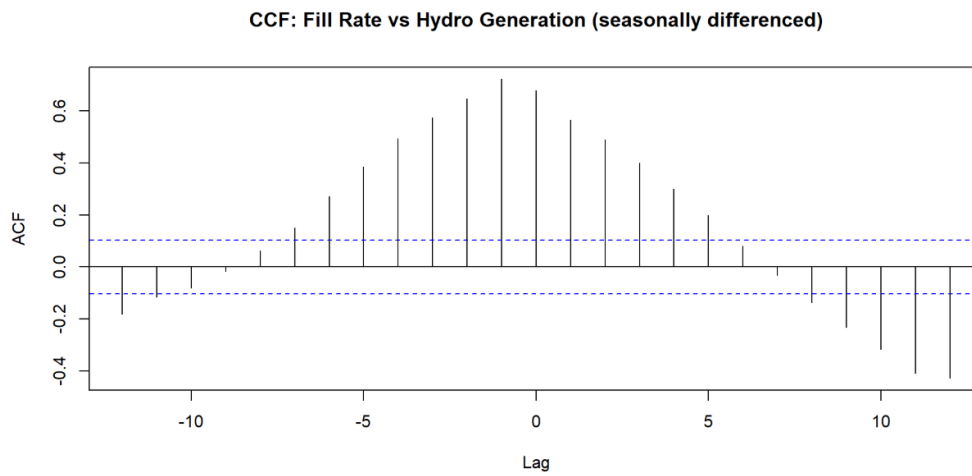


Figure 16: Cross-correlation function (CCF) between seasonally differenced hydro generation and reservoir fill rate.

A.11 Pearson correlation coefficient

```
## Pearson's product-moment correlation
##
## data:  reservoir_fill_rate$fill_rate_pct_lag1 and reservoir_fill_rate$TWh
## t = 9.6268, df = 372, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.3615533 0.5242511
## sample estimates:
##          cor
## 0.4465863
```

Figure 17: Pearson correlation between lagged reservoir fill rate and hydro generation